Manual for KVM Setup via command line on Linux Host

This guide is divided into three main sections: Basic Installation, Enable Paravirtualization, and Manual Hard Disk Partitioning.

Section 1: Basic KVM Installation

Step 1: Check virtualization support

Open the terminal on your Linux host:

bash
egrep -c '(vmx|svm)' /proc/cpuinfo

- `vmx` = Intel virtualization.
- `svm` = AMD virtualization.
- A result greater than `0` means virtualization flags are available.

If the result is `0`, enable virtualization in BIOS/UEFI.

Step 2: Install KVM packages

For Ubuntu/Debian:

bash
sudo apt update
sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients virt-manager ovmf

Step 3: Start the virtualization service

bash
sudo systemctl enable --now libvirtd
sudo usermod -aG libvirt,kvm "$USER"

Log out and log back in.

Step 4: Verify installation

bash
ls -l /dev/kvm

virsh -c qemu:///system list --all

If `/dev/kvm` exists and the `virsh` command works, proceed to VM creation.

Step 5: Create a virtual machine

1. Download an operating system ISO, such as Ubuntu.
2. Open Virtual Machine Manager:
   bash
   virt-manager
3. Click Create a New Virtual Machine.
4. Select Local install media (ISO image).
5. Select the downloaded ISO.
6. Allocate resources, for example:
   - RAM: 4 GB
   - CPUs: 2
   - Virtual disk: 40 GB
7. Create the VM and start the OS installation.

Section 2: Enable Paravirtualization

Paravirtualization allows the guest OS to communicate efficiently with virtual hardware through virtualization-aware drivers. With KVM, use VirtIO devices.

Step 1: Open VM hardware settings

1. Shut down the VM.
2. Open Virtual Machine Manager.
3. Select your VM and click Open.
4. Click Show virtual hardware details.

Step 2: Enable VirtIO disk

1. Select the virtual disk.
2. Set Disk bus to `VirtIO`.
3. Apply the changes.

Step 3: Enable VirtIO network

1. Select the network interface.
2. Set Device model to `virtio`.
3. Apply the changes.

Step 4: Verify paravirtualization

Start the VM and run these commands inside a Linux guest:

```
bash
lspci | grep -i virtio
lsmod | grep virtio
```

If VirtIO devices are detected, the guest is using paravirtualized devices.

Note: Linux usually includes VirtIO drivers by default. Windows guests may require separate VirtIO drivers.

Section 3: Manual Hard Disk Partitioning

This section explains how to manually partition the virtual hard disk inside the VM, not the physical hard disk of the host.

Steps to Create Manual Partitions in a KVM Virtual Machine

For a 40 GB virtual hard disk, create four partitions: EFI, root, swap, and home.

Step 1: Open the OS installer

1. Start your VM in Virtual Machine Manager.
2. Boot from the Linux ISO.
3. Follow the installation instructions until you reach Storage Configuration or Installation Type.
4. Select Manual partitioning, Custom storage layout, or Something else, depending on the Linux distribution.

Step 2: Select the virtual hard disk

- Select the 40 GB virtual disk, usually `/dev/vda` or `/dev/sda`.
- Choose to create a new GPT partition table if the disk is empty.
- Select the available free space to create your partitions.

Important: Make sure you select the VM's virtual disk, not your host's physical hard disk.

Step 3: Create the EFI partition

1. Select the free space and click Add partition (+).

2. Set the size to 512 MB.
3. Choose FAT32 or EFI System Partition.
4. Set the mount point to `/boot/efi`, if requested.
5. Confirm the partition.

Step 4: Create the root partition

1. Select the remaining free space and click Add partition (+).
2. Set the size to 20 GB.
3. Choose the ext4 filesystem.
4. Set the mount point to `/`.
5. Enable formatting for the new partition.

Step 5: Create the swap partition

1. Select the remaining free space and click Add partition (+).
2. Set the size to 4 GB.
3. Select swap area or Linux swap as the partition type.
4. Do not assign a mount point.

Step 6: Create the home partition

1. Select all remaining free space.
2. Click Add partition (+).
3. Choose the ext4 filesystem.
4. Set the mount point to `/home`.
5. Enable formatting for the new partition.

Step 7: Review the partition layout

Your partition table should look approximately like this:

Partition	Size	Filesystem	Mount point
EFI	512 MB	FAT32	`/boot/efi`
Root	20 GB	ext4	`/`
Swap	4 GB	Linux swap	None
Home	Remaining space	ext4	`/home`

Step 8: Apply and install

1. Review the changes carefully.
2. Confirm that only the VM's virtual disk will be modified.
3. Click Done, Apply, or Continue, depending on the installer.
4. Confirm the disk changes.
5. Continue installing Linux.
6. Restart the VM after installation and remove or disconnect the installation ISO.

Step 9: Verify after installation

After Linux boots, open a terminal in the VM and check the Disks application or the filesystem information to verify that:

```
bash
lsblk -f
df -h
```

- `/` is mounted on the root partition.
- `/home` is mounted on the separate home partition.
- `/boot/efi` is mounted on the EFI partition.
- Swap is enabled.

Note: These steps assume a UEFI-based VM. The exact installer labels differ between Ubuntu, Debian, Fedora, and other Linux distributions.

Section 4: Final Verification

Run these commands on the Linux host:

```
bash
virsh -c qemu:///system list --all
virsh -c qemu:///system dominfo YOUR_VM_NAME
```

Replace `YOUR_VM_NAME` with your VM's name.

Verify that:

- KVM is installed and working.
- The virtual machine starts successfully.
- VirtIO disk and network devices are configured.
- The guest OS is installed.
- Manual partitions are mounted correctly.

Remember: KVM provides hardware-assisted virtualization, VirtIO enables paravirtualized devices, and manual partitioning organizes the VM's virtual disk. These are separate steps in configuring your Linux virtual machine.